

JATAYU AIM-X

JATAYU DRONES, MUMBAI

PAYLOAD: *40-42 KG*

ENDURANCE: *50 MINUTES*

- ***Capable of surgical strikes, anti-poaching, anti-piracy operations; burst fire technology***
- ***High-altitude flights***
- ***Multi-weapon integration***
- ***Precision strike, burst fire***

SKYHAWK

IG DRONES, NOIDA

PAYLOAD: *7.5 KG*

ENDURANCE: *3 HOURS*

- ***Vertical take-off & landing***
- ***Security and surveillance and mapping***



HEAVY LIFTING
Left, Satyabrata Sayapathy & Gaurav Achha with their Bon V drone

BON V

BON V TECHNOLOGY, BHUBANESWAR

PAYLOAD: *200 KG*

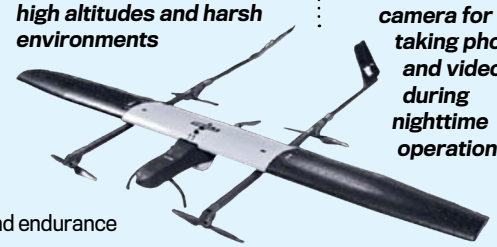
MAX DISTANCE: *40 KM*

- ***India's first AI-powered multi-utility electric air vehicle***
- ***D0311A certified Battery and SiC (silicon carbide)-based high voltage propulsion system ensures a zero-carbon emission model***

SWITCH 1.0

IDEAFORGE, MUMBAI

- ***A fixed-wing drone designed for surveillance missions***
- ***Capable of operating at high altitudes and harsh environments***
- ***Equipped with a 1,280×720 pixel 25x optical zoom daylight camera***
- ***Thermal imaging with 640×480 pixel infrared camera for taking photos and videos during nighttime operations***



Note: Costs of drones range from Rs 5 lakh to Rs 20 lakh, depending on payload and endurance

all-weather terrain and multi-weapon integration for precision strike and burst fire. "Drones are no longer the future in Indian military," says Jatayu founder Amit Johri. "They are the present. The army has been encouraging Indian start-ups as part of emergency procurement of 2,000-plus drones under the Make-2 category.... We are set to become the world's largest military drone fleet."

IG Drones, a three-year-old start-up based in Noida and founded by Odisha Engineering College graduates Bodhisattva Sanghapriya, Om Prakash Das and Shuvam Dash, developed the surveillance drone Skyhawk. The team is now working with the Indo-Tibetan Border Police and the Border Security Force. However, just developing drones is not enough. There must be enough trained personnel to operate them. That is also one of IG Drones' goals. The firm works with over 150 enterprises on drone solutions (for example, crowd monitoring, pipeline inspection, GIS survey, surveillance), training drone pilots and teaching drone technology.

Pence Mataria and his three friends from IIT Kanpur hit the headlines when their company, the Bengaluru-based CDSpace Robotics, developed a drone that delivered blood samples from a remote primary health centre in Uttarakhand's Tehri district to a nearby hospital. The 36 km distance was covered in only 19 minutes.

Now, their sights are set on medium category military surveillance drones, where the market is dominated by Israelis. "Big players do not want to invest in developing machines for the Indian military," says Mataria. He sets great store on their SPOT V3 surveillance drone—which is runway-independent with vertical take-off and landing (VTOL)—for border patrol and maritime surveillance.

While assessing various drones developed by new firms vying for its attention, the army has quietly set upon acquiring drones for immediate use. Early in 2021, it awarded a contract to Delhi-based ideaForge—one of India's largest manufacturer of drones for defence and homeland security—worth $20 million for switch drones, UAVs with VTOL capabilities, which can thereafter transition to regular fixed wing operations. The SWITCH 1.0 from ideaForge is an indigenous fixed-wing surveillance drone capable of operating at high altitudes and in harsh environments. The aircraft is equipped with a 1,280×720 pixel 25x optical zoom daylight camera and a thermal imaging 640×480 pixel infrared camera for taking photos and video during nighttime operations. The army's special forces and infantry are currently using them at Daulat Beg Oldi in Ladakh, close to the Line of Actual Control. Earlier, the infantry had to depend on other teams for surveillance; now, these drones enable them to provide tactical inputs to units deployed on the border. "After the army's aviation and artillery units, the IAF is eyeing them," says Nitin Kaul of ideaForge.

UAV start-ups are scrambling to seize its immense economic potential. Through the crescendo of lift-offs, realisation dawns that we live in the age of drones. ■